

A.1 Test Plan

Performed by: [Minh Thư] **Reviewed by:** [Thiên Phước, Kim Hằng, Khánh Linh, Gia Phúc] **Edited by:** [Minh Thư]

A.1.1 Test Objectives

- Verify that implemented features work according to the requirements.
- Verify that user workflows work correctly from end to end.
- Identify functional defects and edge cases.
- Validate AI-powered features with representative inputs.
- Verify that previously identified bugs have been fixed.

A.1.2 Test Scope

In Scope

The following features are included in functional testing:

- Authentication
- Onboarding
- Personalized Learning Path
- Study Groups
- AI Speaking Assistant

Testing covers the main functional workflows, valid and invalid inputs, error handling authorization, data persistence, and relevant edge cases for these features.

Out of Scope

The following features are not included in the formal functional test suite:

- Flashcards
- Pomodoro
- Group Chat

These features may be manually checked during the final product demonstration but are not included in the 50 formal functional test cases.

A.1.3 Features to Be Tested

Feature ID	Feature	Related Use Case	Priority
F01	Authentication and O	UC01	High
F02	Onboarding	UC02	High
F03	Personalized Learning Path	UC03	High
F04	Study Groups	UC04	High

Feature ID	Feature	Related Use Case	Priority
F05	AI Speaking	UC05	High

A.1.4 Test Environment

Category	Configuration
Operating System	macOS / Windows
Browser	Chrome
Frontend	React + Vite
Backend	NestJS
Database	PostgreSQL / Supabase
Deployment	Vercel / Render
Testing Type	Manual Functional Testing

A.1.5 Testing Tools

- Browser Developer Tools
- Postman / Swagger
- PostgreSQL / Supabase
- Git
- Spec Kit
- [Other tools]

A.1.6 Test Schedule

Phase	Date	Activity	Responsible
Preparation	[18th August]	Prepare test cases	[Minh Thư]
Execution	[19th August]	Execute test cases	[Minh Thư]
Bug Fixing	[20th August]	Fix defects	[Minh Thư]
Regression Testing	[21th August]	Re-test fixed features	[Minh Thư]
Final Review	[22th August]	Review results	[Minh Thư]

A.1.7 Responsibilities

Member	Responsibility
[Minh Thư]	Test planning
[Kim Hằng]	Authentication and onboarding testing
[Thiên Phước]	Personalized Learning Path

Member	Responsibility
[Khánh Linh]	Study group testing
[Gia Phúc]	AI feature testing
[Minh Thư]	Final review

A.1.8 Entry Criteria

Testing can begin when:

- Required features are implemented.
- The application can be launched successfully.
- Test environment is available.
- Test data is prepared.
- Test cases have been reviewed.
- Critical blocking defects from previous testing have been resolved.

A.1.9 Exit Criteria

Testing is considered complete when:

- All planned test cases have been executed.
 - All failed test cases have corresponding bug reports.
 - Critical and high-severity defects are resolved or documented.
 - Regression testing has been completed.
 - Test results have been reviewed by the team.
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